Vidisha Deshpande

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Education

State University of New York, Stony Brook | *Masters in Data Science*

- **GPA: 3.78** Coursework: Natural Language Processing, Data Structures & Algorithms, Data Analysis, Statistics

University of Pune | *Bachelors in Engineering*

- **GPA: 9.23/10** Coursework: Object Oriented Programming, Database Systems, Deep Learning

Experience

BMC Helix *Jun 2026 – Aug 2026*
Data Science Intern

- Built a conversational ITSM agent on BMC Helix Agent Studio that translates unstructured natural-language input into structured tickets, replacing traditional form-based intake with LLM-driven field inference and deterministic priority assignment.
- Prototyped natural-language ticket search using LLM-generated keyword expansion with lightweight text matching.
- Evaluated a production conversational AI agent in an enterprise IT portal through structured adversarial testing, identifying and documenting failure modes in intent routing and multi-turn conversation state handling.

Pune Institute of Computer Technology *github.com/vidi-deshp12*
Research Assistant *Aug 2024 – May 2025*

- Developed a real-time **image captioning system with audio feedback** for visually impaired users, integrating **CLIP–GPT-2** image captioning with EasyOCR-based scene-text extraction to capture both visual and textual context.
- Fine-tuned the **ClipCap** architecture for the accessibility domain using parameter-efficient **prefix tuning** on **20K+** VizWiz image-caption pairs, specializing the model for descriptions of scenes encountered by visually impaired users.
- Optimized inference by parallelizing CLIP image encoding and EasyOCR scene-text extraction, reducing end-to-end latency by **44%** (9s $\rightarrow$ 3.2s) for more responsive real-time use.
- Built a fully functional end-to-end prototype with a **Flask inference API** and **React Native** mobile app, validating performance under low-light, motion blur, and occlusion; co-authored a peer-reviewed Springer LNNS publication based on the work.

Projects

FactGraph: Fact-Checking Layer for LLMs *github.com/vidi-deshp12*
Python, spaCy, Neo4j, LLaMA 3.1, MiniLM, DeBERTa, Qwen2.5 *Mar 2026 – May 2026*

- Built a fact-checking pipeline for LLM-generated claims, combining LLM annotation, information extraction, knowledge-graph retrieval, and NLI verification; reduced annotation errors by **62%** across 7,724 FEVER dev claims.
- Improved NEI recall **4.9×** (0.17 $\rightarrow$ 0.83) by grounding ambiguous claims in structured Wikidata evidence rather than relying on LLM parametric knowledge.
- Engineered a three-route retrieval system (exact match $\rightarrow$ property lookup $\rightarrow$ MiniLM semantic fallback) over a 3,990-triple Wikidata graph, diagnosing schema mismatches and normalization failures across 700 entities.

SciFact: Evidence-Based Scientific Claim Verification *github.com/vidi-deshp12*
Python, FAISS, Hugging Face, sentence-transformers, DeBERTa-v3 *July 2026 – Present*

- Building an evidence-based pipeline to verify scientific claims without relying on LLM memory, retrieving supporting evidence from 5,183 biomedical abstracts with MiniLM embeddings and FAISS; achieved **92.6%** Recall@10.
- Diagnosed 14 retrieval failures across vocabulary, granularity, and document-length mismatches to guide targeted ablations rather than arbitrary model changes.
- Extending the pipeline with sentence-level evidence selection and DeBERTa-v3 NLI verification, enabling end-to-end attribution of failures to retrieval, evidence selection, or classification.

Publications (peer-reviewed)

Empowering Vision: A Survey on Assistive Technologies for the Visually Impaired **(SmartCom 2025)**
Few-Shot Learning in Vision-Language Models: A Survey **(CVR 2025)**

Technical Skills

Languages: Python, SQL
Databases: PostgreSQL, Neo4j, FAISS
ML/DL Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers
Tools & Platforms: Git, Docker, Flask, spaCy, Cursor
Specializations: NLP, LLM Evaluation, Retrieval-Augmented Systems